

Minimal Tree

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Given two integers, **L** and **C**, your task is to implement a function that constructs and returns a tree following these criteria:

- The tree is **0-indexed**.
- The tree must contain **at least C simple paths of exactly length L**.
- The tree must contain the **minimum possible number of nodes** while satisfying the above conditions.

Implementation Details

COMPILED WITH -Werror -Wreturn-type

You must implement the following procedure:

```
std::vector<std::pair<int, int>> createMinimalTree(int L, int C)
```

- This procedure will be called exactly once.
- The returned `std::vector<std::pair<int, int>>` should represent the edges of the tree. Node IDs must use consecutive integers starting from 0 (i.e., 0, 1, 2, ..., n-1, where n is the total number of nodes in your constructed tree). Each `std::pair<int, int> {u, v}` denotes an undirected edge between nodes u and v.

Constraints

- $1 \leq L \leq 200,000$
- $1 \leq C \leq 2,147,483,647$
- For all valid inputs **L** and **C**, the optimal minimal tree is guaranteed to have between 1 and 200,000 nodes (inclusive).

Scoring

Your solution will be evaluated based on the number of nodes **n** in the tree returned by your `createMinimalTree` function, compared to the minimum number of nodes (expected) for the given **L** and **C**. Your final score will be the average score achieved across all test cases.

A submission will receive a **Wrong Answer (WA)** if:

- The returned graph is not a valid tree (e.g., contains cycles, is disconnected, or has an incorrect number of edges for n nodes).
- The returned tree's node IDs are not 0-indexed or do not use consecutive integers (0, 1, ..., $n-1$).
- The returned tree does not contain at least **C** simple paths of exactly length **L**.
- $n > 1,000,000$

If the submission does not result in a Wrong Answer based on the above criteria, it will be scored as follows:

- If $n \leq \text{expected} + 2$: **Accepted (AC)**. (Optimal solution).
- If $n > \text{expected} + 2$: **Partially Correct** ($100 * ((\text{expected}+2)/n)^2 \%$).

In at least 20% of all test data, $L \leq 2$.

Examples

```
createMinimalTree(1, 1)
```

returns `{{0, 1}}`

Sample Grader

The sample grader reads input as follows:

- Line 1: Two integers, **L** and **C**.

The sample grader will print the output from your `createMinimalTree` function in the following format:

- Line 1: The total number of nodes in the constructed tree, n .
- Subsequent lines: Each line contains two integers, u and v , representing an edge in the tree.

Limits

- Time limit: 1 seconds
- Memory limit: 512 MB

NOTE: COMPILED WITH -Werror -Wreturn-type

C++ Template

```
#include <bits/stdc++.h>
using namespace std;

vector<pair<int, int>> createMinimalTree(int l, int c) {
    vector<pair<int, int>> graph;
    // Write your code here

    return graph;
}

// int main() {
//     int l, c;
//     cin >> l >> c;
//     const vector<pair<int, int>> ans = createMinimalTree(l, c);
//     cout << ans.size() + 1 << '\n';
//     for (const auto& [u, v] : ans) cout << u << ' ' << v << '\n';
//     return 0;
// }
```

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